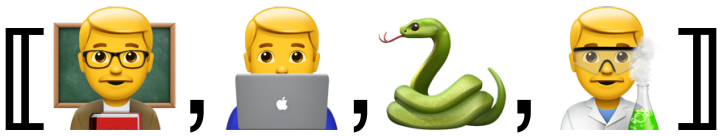
Lecture Notes for

Neural Networks I



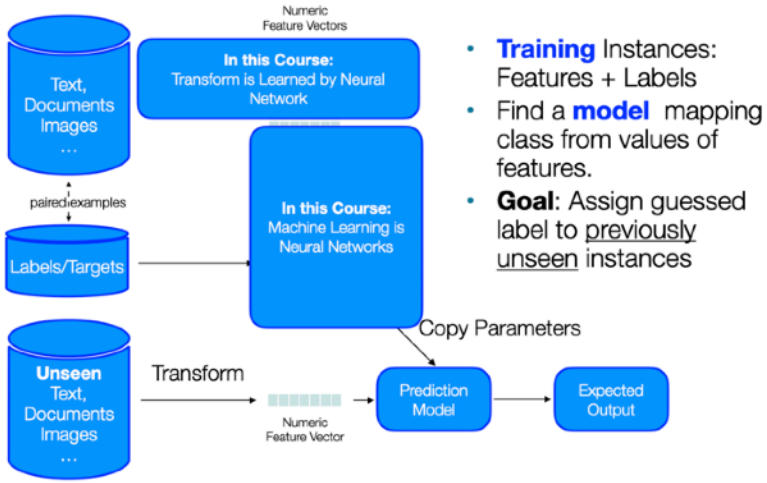
Professor Eric Larson

Table Data with Numpy and Pandas

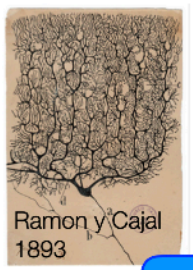
Class Logistics and Agenda

- Canvas. Videos. Attendance. Participation.
- Agenda:
  - Data Encodings
  - Demo: Table Data, Numpy, Pandas
  - Data Quality and Imputation

Last Time

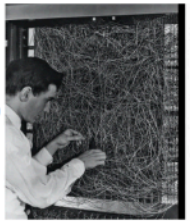


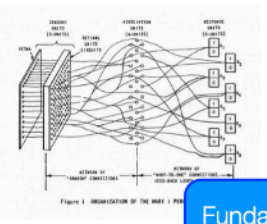
- **Training** Instances: Features + Labels
- Find a **model** mapping class from values of features.
- **Goal**: Assign guessed label to previously unseen instances

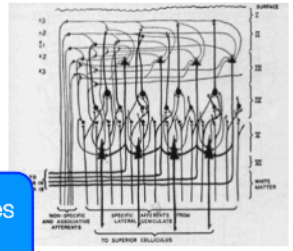


Biologically Inspired

Turns into Optimization of Matrices
...







Fundamentally guides Connectivity

4: Frank Rosenblatt with his Mark I perceptron(left), and a graphical representation of it(right).
https://data-science-blog.com/blog/2020/07/16/a-brief-historyof-neural-nets-everything-you-
before-learning-lstm/

Most Install Instructions (Windows, Linux, and Intel-base Macs)

For operating systems that do NOT use Apple Silicon, you can use the following commands:

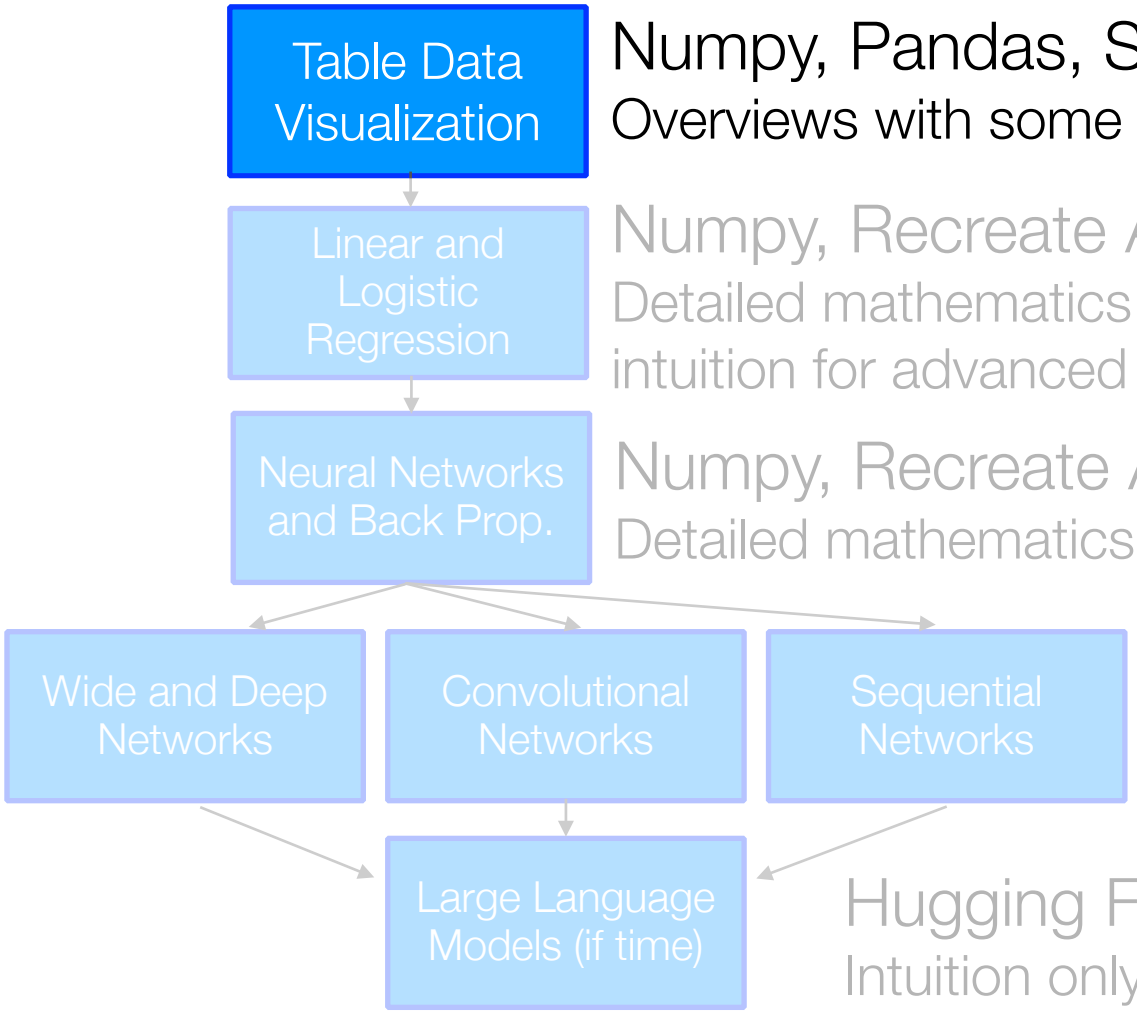
```
conda create --name mlenv3_12 python=3.12
conda activate mlenv3_12
pip install jupyter numpy scipy pandas matplotlib scikit-learn plotly missingno
pip install protobuf tensorflow-datasets importlib-resources
pip install torch torchvision transformers
```

Apple Silicon MacOS Install Instructions

Note: Apple M-series Macs should use a different install procedure as follows:

```
conda create --name mlenv3_12 python=3.12
conda activate mlenv3_12
python -m pip install tensorflow-macos
python -m pip install tensorflow-metal
pip install jupyter pandas matplotlib scikit-learn plotly missingno
pip install "numpy<2.0" "scipy<=1.13.1"
pip install protobuf tensorflow-datasets importlib-resources
pip install torch torchvision transformers
```

Class Overview, by topic



Numpy, Pandas, Seaborn

Overviews with some in-depth discussion

Numpy, Recreate API for Scikit-learn

Detailed mathematics for simple optimization

intuition for advanced optimization

Numpy, Recreate API for DL Frameworks

Detailed mathematics for NN operations

Keras, Tensorflow, Torch

Intuition, Detailed implement.

Hugging Face

Intuition only

Types of Data and Categorization

Table Data

- **Table Data**: Collection of data **instances** and their **features**
  - **Python:** Pandas Dataframe
  - **R:** Data.frame
  - **Matlab:** Table Class
  - **C++**: Make your own,

```
std::vector<Record>
```

Self Test: Table data storage is not memory efficient.

a) True
b) False
c) It depends on the backend
d) It depends on the programming language of the interface

Attributes, columns, variables, fields, characteristics, **Features**

Target, Class, Label

Objects, records, rows, **points**, **samples**, cases, entities, **instances**

	Pregnant	*BMI*	*Age*	*Diabetes*
1	Y	33.6	41-50	positive
2	N	26.6	31-40	negative
3	Y	23.3	31-40	positive
4	N	28.1	21-30	negative
5	N	43.1	31-40	positive
6	Y	25.6	21-30	negative
7	Y	31.0	21-30	positive
8	Y	35.3	21-30	negative
9	N	30.5	51-60	positive
10	Y	37.6	51-60	positive

Feature Vector Representation

$f_{mono}(\cdot)$ monotonic function

Discrete

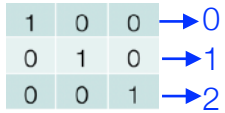
Categorical or Nominal

Variable could be one value in a set of categories. No ordering.

Example: Employee ID

Allowed Transforms: permuting values

boolean, one hot encoding, or hash (as integer)



Ordinal

Variable could be one value in a set of categories. Ordering matters.

Example: Star Ratings, 1-5

Allowed Transforms:

$$V_{new} = f_{mono}\left(V_{old}\right) + b$$

integer (or boolean)

Continuous

Interval or Numeric

Value is continuous numeric value. Could be in specified range.

Example: BMI, Temperature, etc.

Allowed Transforms:

$$V_{new} = f_{mono}\left(V_{old}\right) + b$$

float

Ratio or Numeric

Value is continuous numeric value. Zero is meaningful. Often not treated differently than interval.

Example: Length, Elevation

Allowed Transforms:

$$V_{new} = f_{mono}\left(V_{old}\right)$$

float

Data Tables as Variable Representations

Table

TID	Pregnant	BMI	Age	Eye Color	Diabetes
1	Y	33.6	41-50	brown	positive
2	N	26.6	31-40	hazel	negative
3	Y	23.3	31-40	blue	positive
4	N	28.1	21-30	brown	inconclusive
5	N	43.1	31-40	blue	positive
6	Y	25.6	21-30	hazel	negative

Internal Rep.

TID
1
2
3
4
5
6

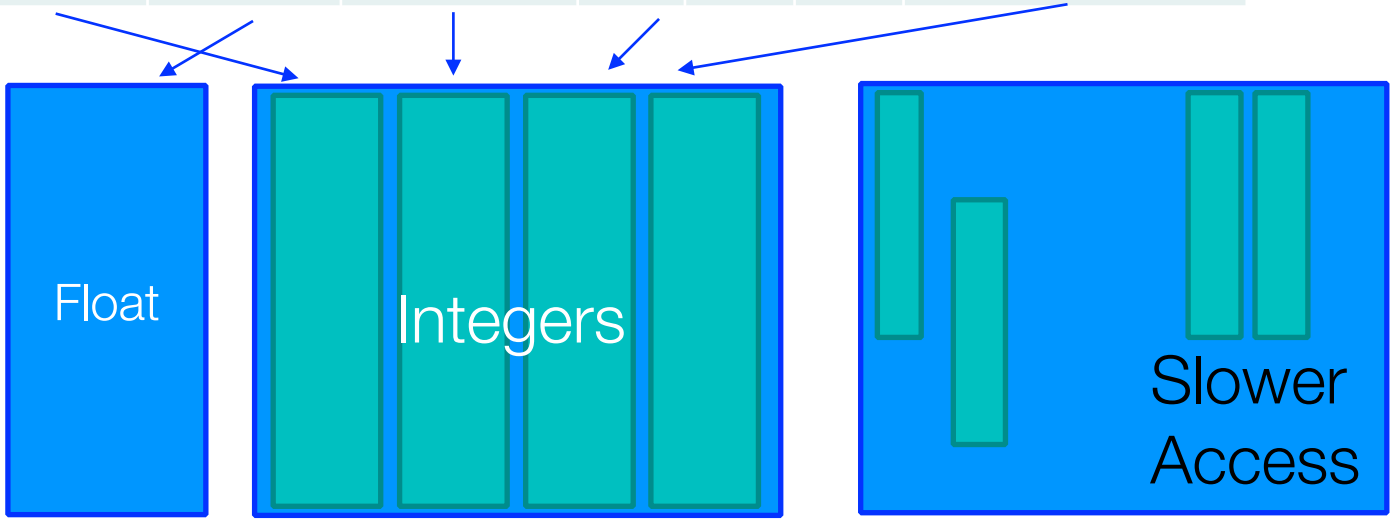
Data Tables as Variable Representations

Internal Rep.

TID	Binary	Float	Ordinal	One Hot			Diabetes
				brown	hazel	blue	
1	1	33.6	2	1	0	0	1
2	0	26.6	1	0	1	0	0
3	1	23.3	1	0	0	1	1
4	0	28.1	0	1	0	0	2
5	0	43.1	1	0	0	1	1
6	1	25.6	0	0	1	0	0

Memory

Fast access via contiguous memory blocks, ideally



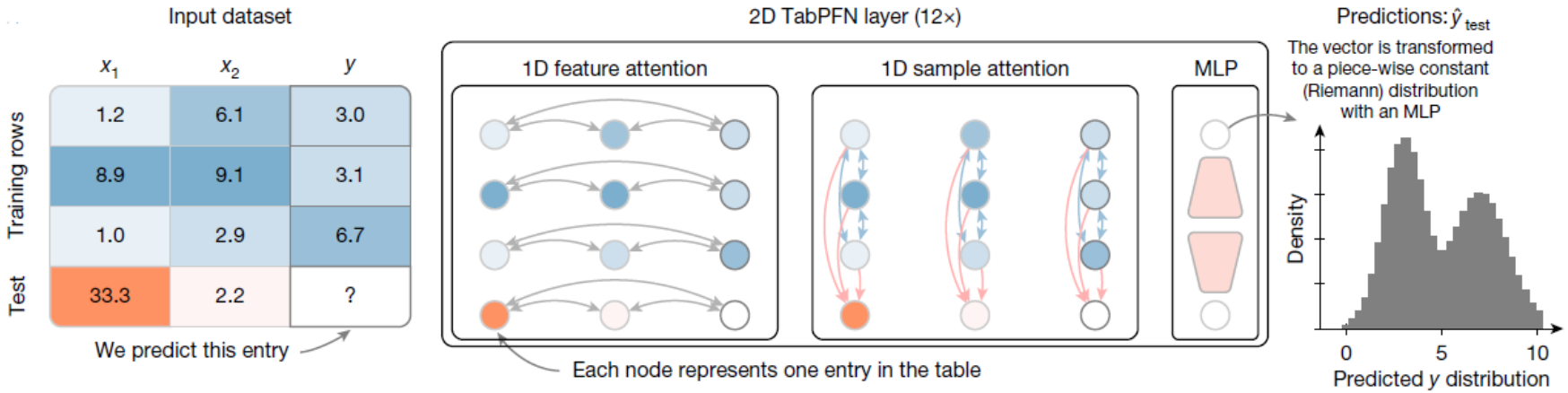
"Finish"
Jupyter Notebooks

01. Numpy and Pandas Intro.ipynb

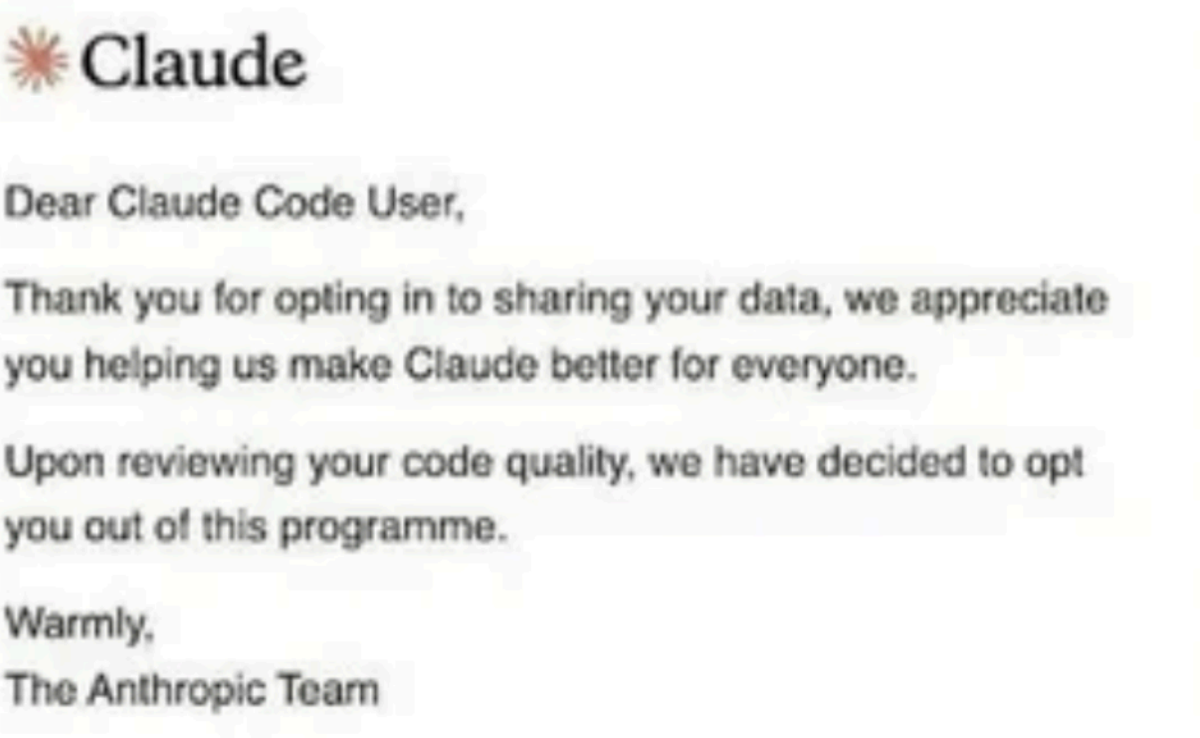
Foundational Models

- For many types of data, foundational models:
  - Pre-trained on large datasets
  - Used as starting points for adapting to new data
  - Efficient and straightforward for neural networks
  - **Text**: BERT, GPT, Llama
  - **Images**: ResNet, ConvNext, Swin, ViT, … YOLO

TabPFN: Foundational Model for Table Data …



Data Quality



Claude

Dear Claude Code User,

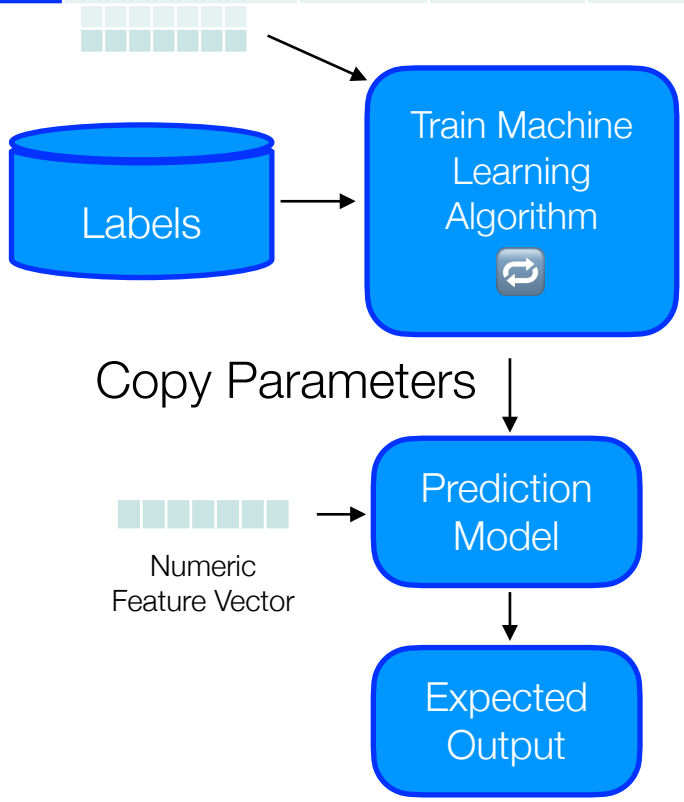
Thank you for opting in to sharing your data, we appreciate you helping us make Claude better for everyone.

Upon reviewing your code quality, we have decided to opt you out of this programme.

Warmly,
The Anthropic Team

Data Quality Problems

TID	Hair Color	Hgt.	Age	Arrested
1	Brown	5'2"	23	no
2	Hazel	1.5m	12	
3	NaN	5	999	no
4	Brown	5'2"	23	no



- Missing
  - Easy to find, NaNs
- Duplicated
  - Easy to find, hard to verify
- Noise or Outlier
  - Hard to define / catch

Information is not collected (e.g., people decline to give their age and weight)

Features **not applicable** (e.g., annual income for children)

UCI ML Repository: 90% of repositories have missing data

Handling Issues with Data Quality

- **Eliminate** Instance or Feature
- **Ignore** the Missing Value During Analysis Replace with all possible values (talk about later)
- **Impute** Missing Values

How?

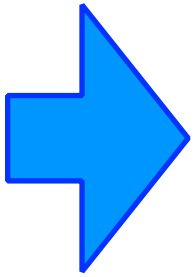
Stats?
mean
median
mode

Imputation

- When is it probably fine to impute missing data:
  - (A) When there is not much missing data
  - (B) When the missing feature is mostly predictable from other features
  - (C) When there is not much missing data for some subgroup of the data
  - (D) When it is the class you want to predict

Split-Impute-Combine

TID	Pregnant	BMI	Age	Diabetes
1	Y	33.6	31-40	positive
2	N	26.6	31-40	negative
3	Y	23.3	?	positive
4	N	28.1	21-30	negative
5	N	43.1	31-40	positive
6	Y	25.6	21-30	negative
7	Y	31.0	21-30	positive
8	Y	35.3	?	negative
9	N	30.5	51-60	positive
10	Y	37.6	21-30	positive



split: pregnant
split: BMI > 32

TID	Pregnant	BMI	Age	Diabetes
1	Y	>32	31-40	positive
8	Y	>32	?	negative
10	Y	>32	21-30	positive

Mode: none, can't impute

TID	Pregnant	BMI	Age	Diabetes
3	Y	<32	?	positive
6	Y	<32	21-30	negative
7	Y	<32	21-30	positive

Mode: 21-30

Try to group similar rows together first, then impute within the group